ITCS 254: Introduction to Linux, Shell, and Commands

Lab 10 – Manipulating Text

1. [From Lab9] Create an empty file named test.txt and check its permissions.

```
root@ubuntu:~$ touch test.txt
root@ubuntu:~$ ls -la test.txt
-rw-r--r-- 1 root root 0 Oct 12 09:10 test.txt
```

2. [From Lab9] Change bob to be the owner and dev2 to be the group owner of test.txt

```
root@ubuntu:~$ chown bob test.txt
root@ubuntu:~$ ls -l test.txt
-rw-r--r-- 1 bob root 0 Oct 12 09:10 test.txt
root@ubuntu:~$ chgrp dev2 test.txt
root@ubuntu:~$ ls -l test.txt
-rw-r--r-- 1 bob dev2 0 Oct 12 09:10 test.txt
```

3. [From Lab9] Change permission of test.txt to be `rwxrw-r--` by numeric mode and display output.

```
root@ubuntu:~$ chmod 764 test.txt
root@ubuntu:~$ ls -l test.txt
-rwxrw-r-- 1 bob dev2 0 Oct 12 09:10 test.txt
```

4. [From Lab9] Change permission of test.txt to be `rw-rw-rw-` by symbolic mode and display output.

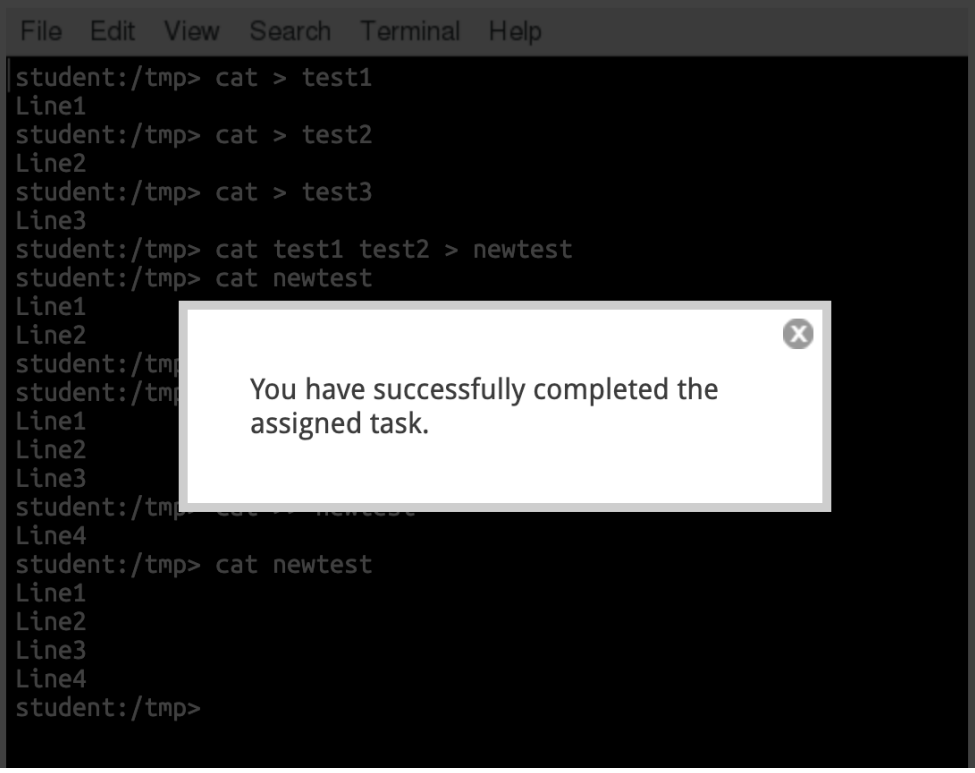
```
root@ubuntu:~$ chmod 666 test.txt
root@ubuntu:~$ ls -l test.txt
-rw-rw-rw- 1 bob dev2 0 Oct 12 09:10 test.txt
```

5. Cat practices: http://linuxfoundation.s3-website-us-east-1.amazonaws.com/TIY/usingcat/index.html

Try-It-Yourself: Using `cat`

Tasks to be performed:

1. Using `cat`, create a file named test1, type "Line1" and save it by pressing Ctrl-D.
2. Using `cat`, create a file named test2, type "Line2" and save it by pressing Ctrl-D.
3. Using `cat`, create a file named test3, type "Line3" and save it by pressing Ctrl-D.
4. Concatenate the files test1 and test2 into a file named newtest.
5. View the contents of newtest, using `cat`.
6. Append the file test3 at the end newtest.
7. View the contents of the newtest file, using `cat`.
8. Append the following text to newtest: "Line4".
9. View the contents of the newtest file, using `cat`.



6. Using sed
   a. Create a text file containing:
      This is a first line
      This is a second line
      This is a third line
   b. Use sed command to replace “is” with “was” in line 1 – 2.

```
root@ubuntu:~$ sed -e 1,2s/is/are/g test.txt
Thare are a first line
Thare are a second line
This is a third line
```

   c. Use sed command to replace “a” with “the” for the whole text file and save it to another file.

```
root@ubuntu:~$ sed -i s/a/for/g test.txt
root@ubuntu:~$ cat test.txt
Thforre forre the first line
Thforre forre the second line
This is the third line
```

7. Search for all instances of the user command interpreter (shell) equal to /sbin/nologin in /etc/passwd and replace them with /bin/bash (do not overwrite /etc/passwd).

```
root@ubuntu:~$ sed -e 's/\/sbin\/nologin/\/bin\/bash/g' /etc/passwd
```

8. Do the following
   a. Create a text file named planets containing:
      Mercury
      Mars
      Earth
      Neptune
      Earth
      Venus
      Mercury
   b. Count the number of lines in the file

```
root@ubuntu:~$ cat > planets.txt
Mercury
Mars
Earth
Neptune
Earth
Venus
Mercury
root@ubuntu:~$ wc -l planets.txt
7 planets.txt
```

c. Sort the planets file

```
root@ubuntu:~$ sort planets.txt
Earth
Earth
Mars
Mercury
Mercury
Neptune
Venus
```

d. Remove duplicate entries from the planets file

```
root@ubuntu:~$ sort -u planets.txt
Earth
Mars
Mercury
Neptune
Venus
```

9. Grep practice

a. Search for your username in file /etc/passwd.

```
root@ubuntu:~$ cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash
```

b. Find all entries in /etc/services that include the string ftp.

```
root@ubuntu:~$ cat /etc/services | grep ftp
ftp-data        20/tcp
ftp             21/tcp
tftp            69/udp
ftps-data       989/tcp                 # FTP over SSL (data)
ftps            990/tcp
venus-se        2431/udp                # udp sftp side effect
codasrv-se      2433/udp                # udp sftp side effect
gsiftp          2811/tcp
zope-ftp        8021/tcp                # zope management by ftp
```

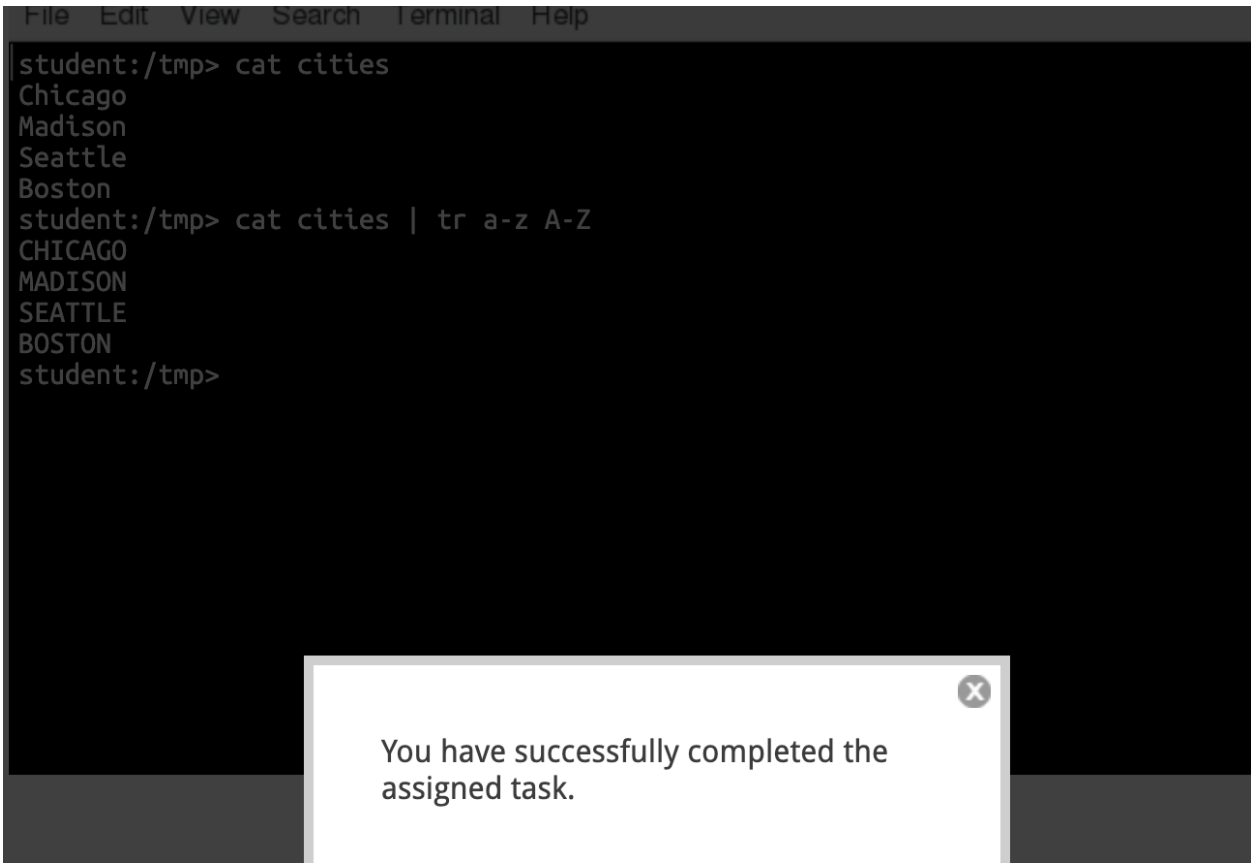
c. Restrict to those that use the tcp protocol.

```
root@ubuntu:~$ cat /etc/services | grep ftp | grep tcp
ftp-data        20/tcp
ftp             21/tcp
ftps-data       989/tcp                 # FTP over SSL (data)
ftps            990/tcp
gsiftp          2811/tcp
zope-ftp        8021/tcp                # zope management by ftp
```

d. Get all strings that start with ts or end with st (use -e option for regular expression)

```
root@ubuntu:~$ grep -e ^ts -e st$ /etc/services
```

10. Tr: http://linuxfoundation.s3-website-us-east-1.amazonaws.com/TIY/usingtr/index.html



11. Extract the first column from /etc/passwd.

```
root@ubuntu:~$ awk -F: '{ print $1 }' /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
systemd-timesync
dhcpcd
messagebus
syslog
systemd-resolve
uuidd
tss
sshd
pollinate
tcpdump
landscape
fwupd-refresh
polkitd
ubuntu
dnsmasq
```